

Time-Varying Sector Betas Across Market Regimes

STAT5261 Final Project Report

Group 22

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Abstract

This report examines whether sector beta should be treated as a stable measure of market sensitivity or as a time-varying exposure that changes across market regimes. Using daily and weekly data for major U.S. sector ETFs from 2018 to 2024, we estimate static CAPM betas, 52-week rolling CAPM betas, and a 26-week robustness specification. The analysis shows that sector beta varies substantially over time and that this variation is economically meaningful. Technology and consumer discretionary generally exhibit higher market sensitivity, while consumer staples, health care, and utilities are more defensive on average. However, the defensive classification is not fixed. During certain regimes, especially the 2022 to 2023 tightening cycle, normally defensive sectors become more exposed to broad market movements. A predictive extension further shows that end-of-month rolling beta is positively associated with next-month realized volatility in the pooled sample, although the strength of this relationship differs by sector. Overall, the results support a dynamic view of systematic risk: rolling beta provides a richer description of sector risk than a single full-sample beta estimate.

1. Introduction

1.1 Motivation

Beta is one of the most widely used measures of systematic risk in empirical finance. In the Capital Asset Pricing Model, beta measures how strongly an asset or portfolio co-moves with the market portfolio after controlling for the risk-free rate. A beta above one indicates that the asset tends to move more than the market, while a beta below one indicates lower market sensitivity. In standard applications, beta is often reported as a single full-sample estimate. This simplification is convenient, but it may hide important changes in risk exposure over time.

This project studies whether sector betas remain stable or vary across different market regimes. The sample period from 2018 through 2024 includes several distinct market environments: the COVID-19 crash, the pandemic recovery, the 2022 bear market, the Federal Reserve rate-hike cycle, the 2023 regional banking crisis, and the 2024 AI-driven technology rally. These episodes provide useful variation for testing whether sector-level market exposure is constant or regime dependent.

The question is economically important for portfolio construction, risk management, and sector rotation. If sector betas rise during crises, diversification may weaken exactly when investors need it most. If technology beta rises during an AI-led rally, the market portfolio may become more concentrated in a small number of firms and themes. If utilities or consumer staples lose their defensive behavior during rate-hike periods, investors relying on static sector labels may underestimate risk. Rolling beta analysis allows us to observe these changes directly.

1.2 Research Questions

- Do sector betas vary meaningfully over time, or are static CAPM betas sufficient?
- Which sectors experience the largest beta changes during major market regimes?
- Do defensive sectors maintain lower and more stable betas during stress periods?
- Can rolling beta help explain or predict near-term realized sector volatility?

Together, these questions evaluate whether rolling beta provides a more informative measure of sector-level systematic risk than a single full-sample estimate.

2. Data and Empirical Design

2.1 Sector ETF and Market Data

The analysis uses major U.S. sector ETFs and SPY as the market benchmark. The sector ETFs are XLK for Technology, XLE for Energy, XLF for Financials, XLV for Health Care, XLP for Consumer Staples, XLU for

Utilities, and XLY for Consumer Discretionary. Daily adjusted close prices are collected from Yahoo Finance from January 1, 2018 to December 31, 2024. Adjusted close prices are used because they incorporate dividends, splits, and other corporate actions, which makes them more appropriate for return calculations.

The selected sectors represent different economic exposures. Technology and consumer discretionary are typically more cyclical and growth-sensitive. Energy is exposed to commodity and supply-demand shocks. Financials are sensitive to credit conditions, rates, and financial stress. Consumer staples, health care, and utilities are commonly treated as defensive sectors. Comparing these groups allows us to test whether conventional sector classifications are supported by the data.

2.2 Macroeconomic and Regime Data

Macroeconomic variables are used to provide regime context rather than to expand the CAPM into a multi-factor model. The 3-month Treasury bill rate is used to construct excess returns. The Federal Funds Rate, 10-year Treasury yield, CPI, VIX, and NBER recession indicator are used to interpret beta dynamics across macroeconomic environments.

Variable	Source & Series ID / Ticker	Purpose
3-month Treasury bill rate	FRED DTB3	Risk-free rate for excess return construction
Federal Funds Rate	FRED FEDFUNDS	Monetary policy environment
10-year Treasury yield	FRED DGS10	Long-term rate environment
CPI	FRED CPIAUCSL	Inflation regime context
VIX	Yahoo Finance ^VIX	Market volatility and risk sentiment
NBER recession indicator	NBER / manual file	Recession shading and event context

Table 1. Macroeconomic and regime variables used in the analysis.

The event windows are designed to capture major market regimes rather than mechanically define all market states. Some windows overlap. When assigning a single regime label to each trading day, more specific events are given priority over broader regimes. For example, the regional banking crisis is treated separately from the broader rate-hike cycle.

Event	Window	Interpretation
COVID crash	2020-02-19 to 2020-03-23	Extreme pandemic-driven selloff
COVID recovery	2020-03-24 to 2020-12-31	Post-crash rebound and policy support
2022 bear market	2022-01-03 to 2022-10-12	Inflation-driven equity stress
Fed rate-hike cycle	2022-03-16 to 2023-07-26	Monetary tightening and rate sensitivity
Regional bank crisis	2023-03-08 to 2023-05-01	Banking stress and financial contagion risk
AI / technology rally	2023-10-01 to 2024-12-31	Technology-led market concentration

Table 2. Event windows used for regime interpretation.

2.3 Data Cleaning and Return Construction

All price and macroeconomic series are aligned to the SPY trading calendar. Daily financial variables are forward-filled where appropriate to account for holidays and missing release dates. Monthly CPI is forward-filled as a step function because inflation is reported discretely rather than continuously. The cleaned daily panel contains aligned prices, returns, excess returns, macro variables, and regime labels. A weekly version of the panel is also created for rolling beta estimation.

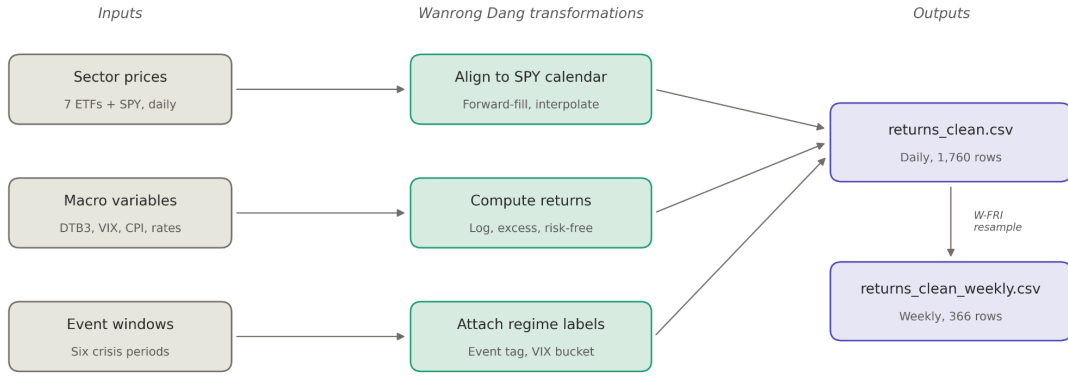


Figure: Wanrong Dang data pipeline. Sector prices and macro variables align to the SPY trading calendar, are combined into log returns and excess returns, and tagged with regime labels.

Figure 1. Data pipeline for sector prices, macro variables, event labels, daily returns, and weekly aggregation.

Log returns are computed as $r_{i,t} = \log(P_{i,t} / P_{i,t-1})$. The daily risk-free rate is constructed from the annualized 3-month Treasury bill rate as $rf_t = DTB3_t / 100 / 252$. Excess returns are then defined as $x_{i,t} = r_{i,t} - rf_t$ for each sector and $x_{m,t} = r_{SPY,t} - rf_t$ for the market benchmark.

Weekly returns are constructed by summing daily log returns within each Friday-ending week. The weekly panel is the primary input for the 52-week rolling beta estimation. This choice reduces high-frequency noise and makes the rolling window approximately equal to one year of market history.

The cleaned daily panel contains 34 columns: 8 price series, 8 log return series, 8 excess return series, the daily risk-free rate, 6 macroeconomic variables, and 2 regime labels (event tag and VIX bucket). The weekly panel uses the same column structure aggregated to Friday-ending weeks.

2.4 Empirical Strategy

The empirical analysis proceeds in five steps. First, we compute descriptive statistics for each sector, including annualized return, annualized volatility, Sharpe ratio, maximum drawdown, skewness, excess kurtosis, and static CAPM beta. Second, we estimate a full-sample static beta for each sector as a baseline. Third, we estimate 52-week rolling CAPM betas to measure time-varying market exposure. Fourth, we compare beta behavior across major market regimes using visual and distributional evidence. Fifth, we test whether end-of-month rolling beta predicts next-month realized volatility.

3. Descriptive Analysis

3.1 Summary Statistics

Table 3 reports the main risk and return characteristics for the seven sector ETFs. XLK has the highest annualized return and the strongest Sharpe ratio, consistent with the technology sector leading the sample period. XLE has the highest annualized volatility and the largest maximum drawdown, reflecting large commodity and energy shocks, especially around the COVID period. Defensive sectors such as XLP and XLV have lower volatility and more contained drawdowns, although their returns are also more moderate.

Sector	Annualized Return	Annualized Volatility	Sharpe Ratio	Max Drawdown	Skewness	Excess Kurtosis	Static Beta
XLK	0.1948	0.2618	0.6543	-0.3356	-0.438	7.7983	1.2506
XLE	0.0654	0.3293	0.1273	-0.6681	-0.9114	15.2633	1.0435
XLF	0.0981	0.2427	0.3075	-0.4286	-0.5872	14.9296	1.0577
XLV	0.0879	0.1761	0.3656	-0.284	-0.4432	10.7165	0.7506
XLP	0.0738	0.1588	0.317	-0.2451	-0.5111	15.8658	0.5999
XLU	0.0855	0.2098	0.2957	-0.3607	-0.217	15.5082	0.642
XLY	0.1264	0.2391	0.4306	-0.3967	-0.7568	7.635	1.104

Table 3. Summary statistics for sector ETF returns, 2018 to 2024.

The static beta estimates support the usual cyclical versus defensive classification. XLK and XLY have betas above one, indicating stronger sensitivity to the market benchmark. XLP, XLU, and XLV have betas well below one, indicating lower full-sample market sensitivity. At the same time, all sectors show negative skewness and high excess kurtosis, which means that return distributions are non-normal and exposed to downside tail risk. This motivates a closer look at stress periods and time-varying beta.

3.2 Sector Performance and Co-movement

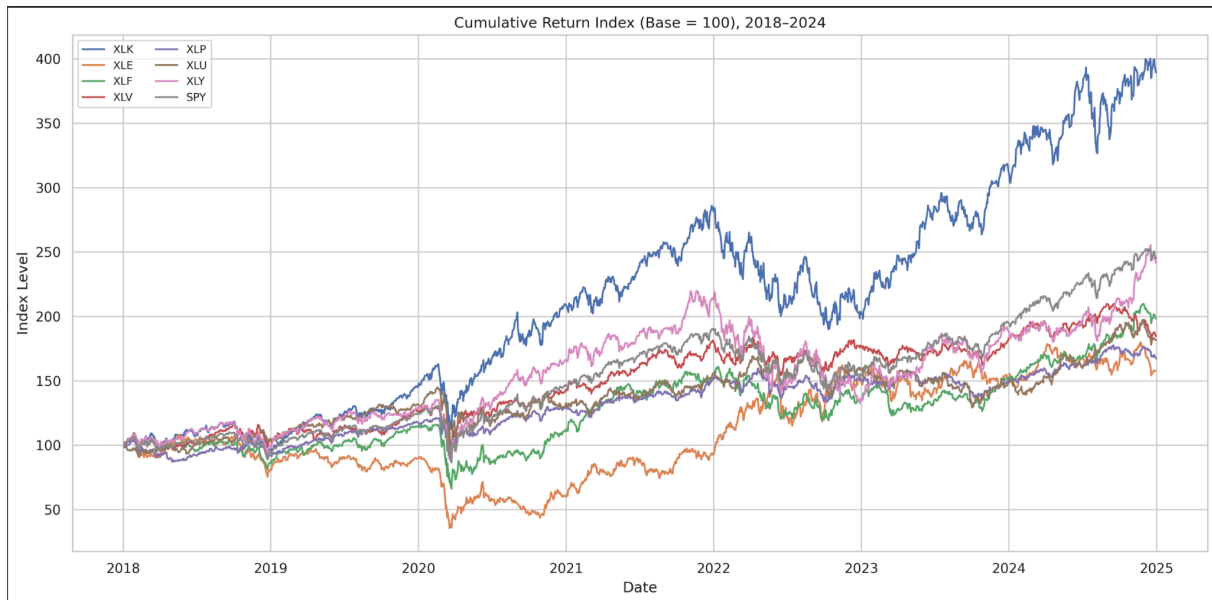


Figure 2. Cumulative return index for SPY and sector ETFs, 2018 to 2024.

The cumulative return index shows that XLK substantially outperformed the other sectors over the sample period, especially after the technology-led rally in 2023 and 2024. XLE experienced the sharpest decline during the COVID crash and remained more unstable than most sectors. Defensive sectors such as XLP, XLV, and XLU followed smoother paths, consistent with their lower volatility and lower beta estimates.

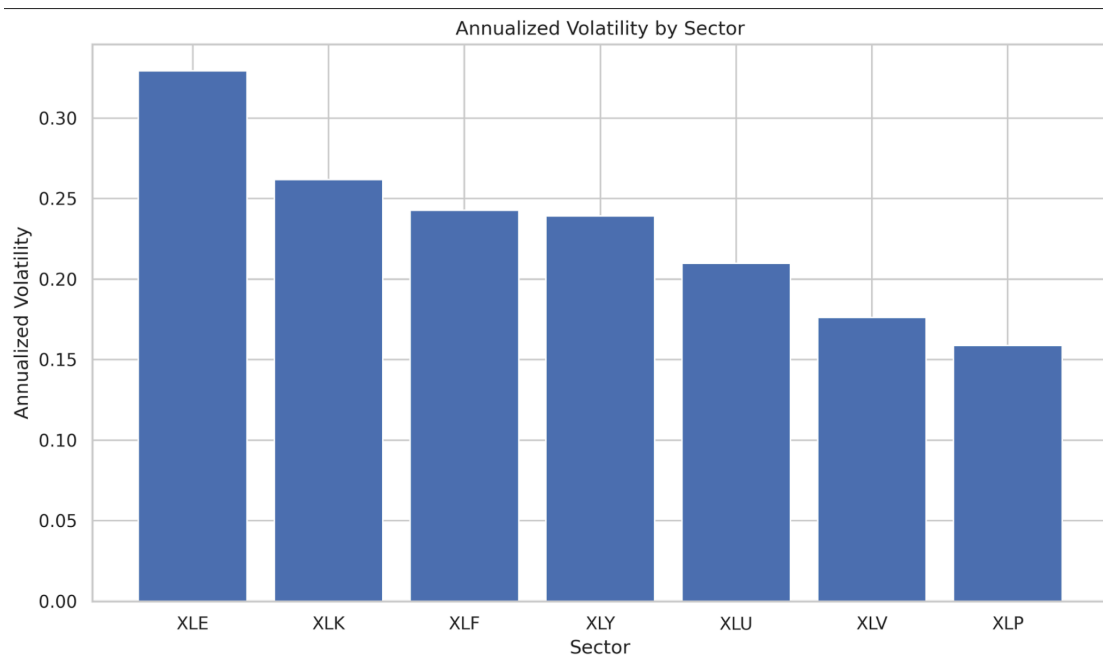


Figure 3. Annualized volatility by sector.

The volatility ranking reinforces the same pattern. XLE has the highest annualized volatility, followed by XLK, XLF, and XLY. XLP and XLV have the lowest volatility, consistent with their role as stabilizing sectors in diversified portfolios.

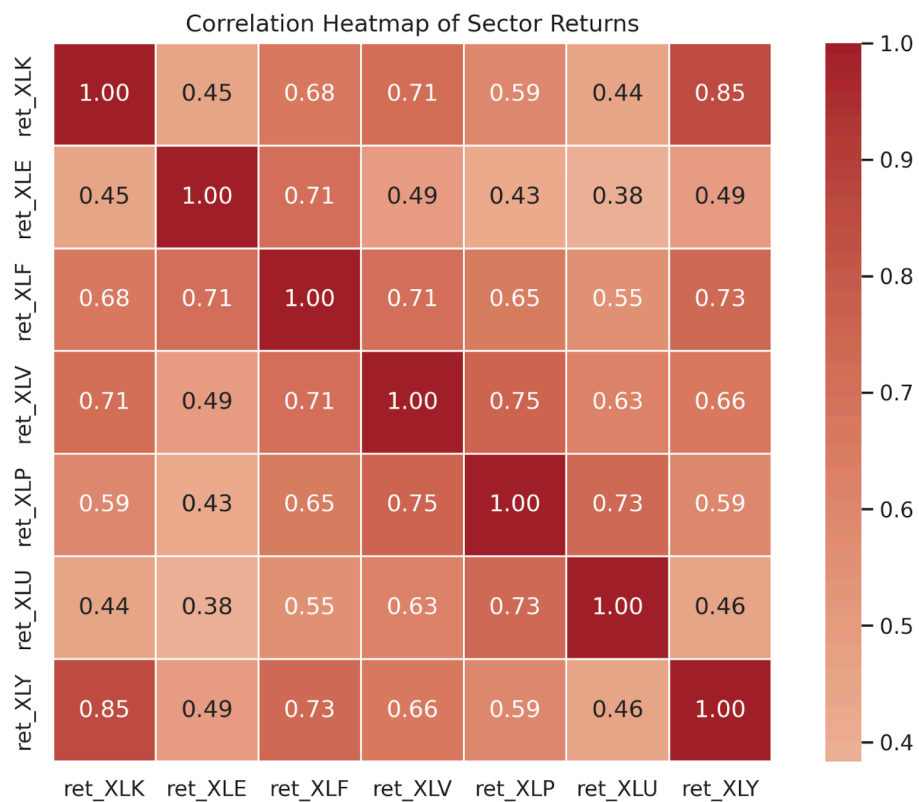


Figure 4. Correlation heatmap of sector returns.

The correlation heatmap shows that all sectors are positively correlated because they share common market-wide drivers. However, the magnitude of co-movement differs. Growth-oriented sectors such as XLK and XLY exhibit strong correlation, while XLU shows lower correlation with several sectors. This suggests that utilities may still provide diversification value, although the strength of that value can change by regime.

4. Rolling Beta Estimation

4.1 Model Specification

For each sector, we estimate a single-factor CAPM regression using excess weekly returns:

$$x_{i,t} = \alpha_i + \beta_i x_{m,t} + \epsilon_{i,t}$$

Here $x_{i,t}$ is the excess return of sector i in week t , and $x_{m,t}$ is the excess return of SPY in the same week. The coefficient β_i measures the sector's systematic exposure to market movements. Instead of estimating one beta for the full sample, we estimate the model repeatedly using a 52-week rolling window. For each window and sector, we record beta, alpha, R-squared, and the standard error of beta.

The 52-week window is the main specification because it balances responsiveness and stability. It is long enough to smooth short-term noise, but short enough to capture changes in market regimes. A 26-week rolling window is used as a robustness check for representative sectors.

4.2 Rolling Beta Results

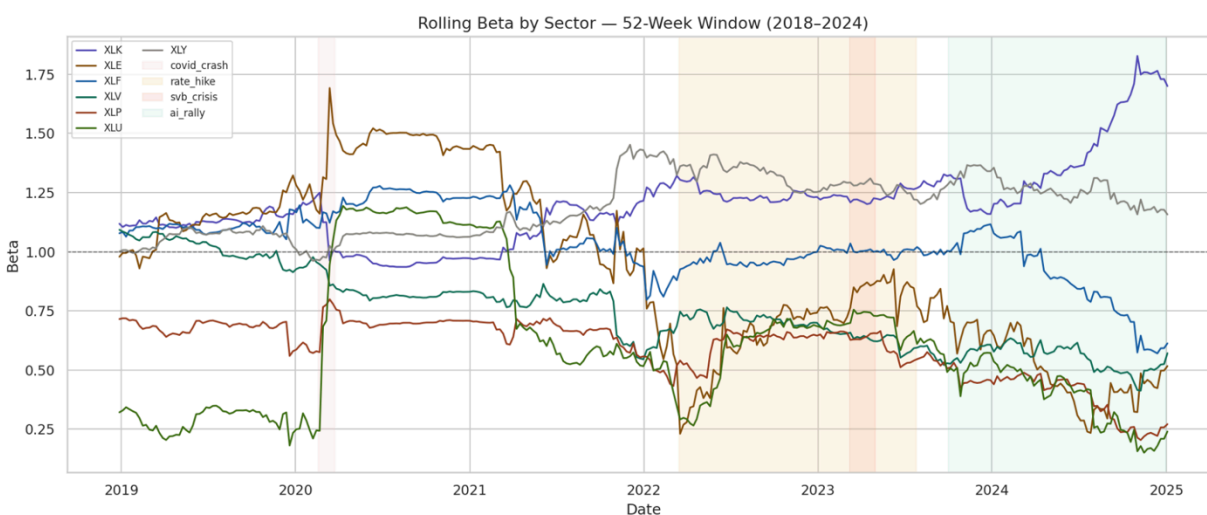


Figure 5. 52-week rolling beta by sector, 2018 to 2024.

Figure 5 shows that sector betas are clearly time-varying. The sectors separate into persistent high-beta and low-beta groups, but the distance between these groups changes across regimes. XLK and XLY generally remain above one, consistent with cyclical and growth-sensitive behavior. XLP, XLU, and XLV generally remain below one, consistent with defensive behavior. However, this classification is not perfectly stable, especially during periods of monetary tightening and market stress.

During the COVID crash, energy beta increased sharply as the sector absorbed a severe demand shock. During the 2022 to 2023 tightening cycle, utilities became more sensitive to the market, consistent with the rate-sensitive nature of bond-like sectors. During the 2024 technology rally, XLK beta rose strongly, reflecting the growing role of large technology firms in driving overall market performance.

These results show why static beta can be misleading. A single full-sample beta compresses several different market environments into one number. Rolling beta instead reveals when and how sector exposures change.

4.3 Robustness to Window Length

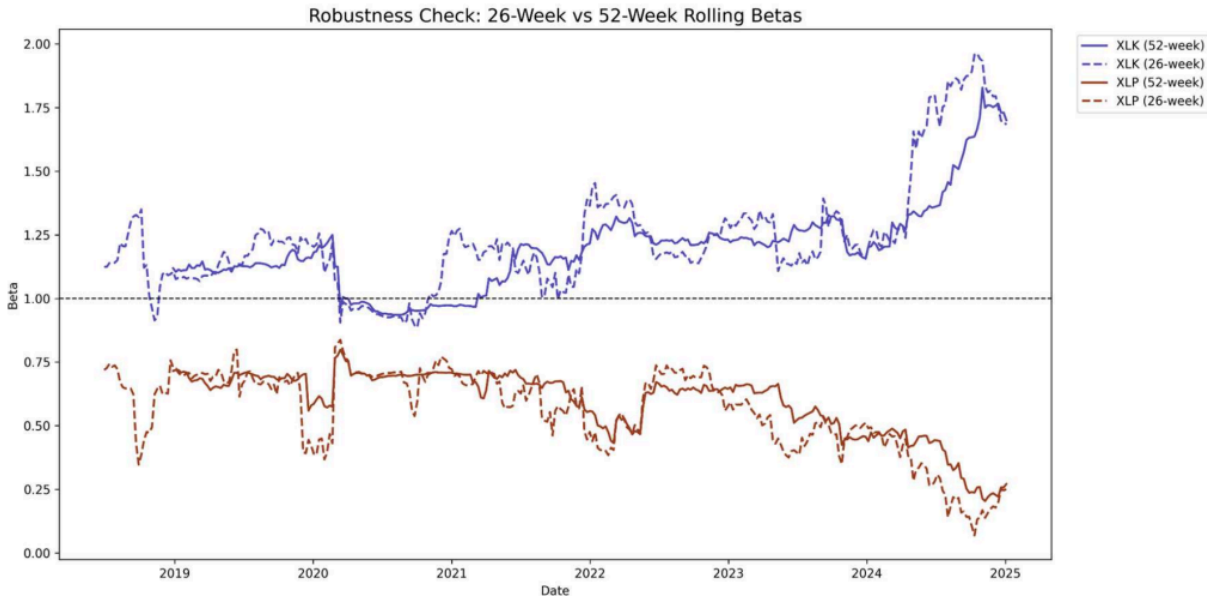


Figure 6. Robustness check using 26-week and 52-week rolling beta for XLK and XLP.

The 26-week window responds more quickly to short-term market movements and therefore produces a noisier series. However, the main conclusion remains unchanged. XLK remains persistently more market sensitive than XLP, and both series display meaningful time variation. The robustness check suggests that the main findings are not driven by a single arbitrary window choice.

5. Regime and Crisis Analysis

5.1 Heatmap Evidence

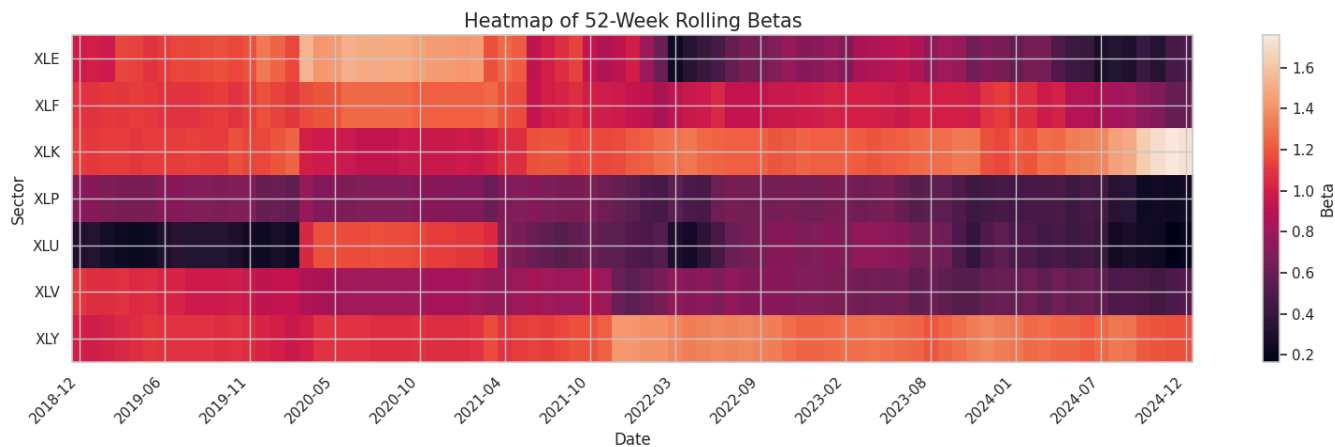


Figure 7. Heatmap of 52-week rolling beta across sectors.

The heatmap summarizes the evolution of beta across both sectors and time. Darker regions indicate higher beta values and stronger market sensitivity. The figure highlights two features. First, sector differences are persistent: technology and consumer discretionary remain high-beta sectors for much of the sample, while staples and utilities remain lower-beta sectors. Second, beta changes over time in response to macroeconomic regimes. The COVID crash, the rate-hike cycle, and the AI-driven rally all coincide with visible shifts in cross-sector beta intensity.

5.2 Distribution of Rolling Beta

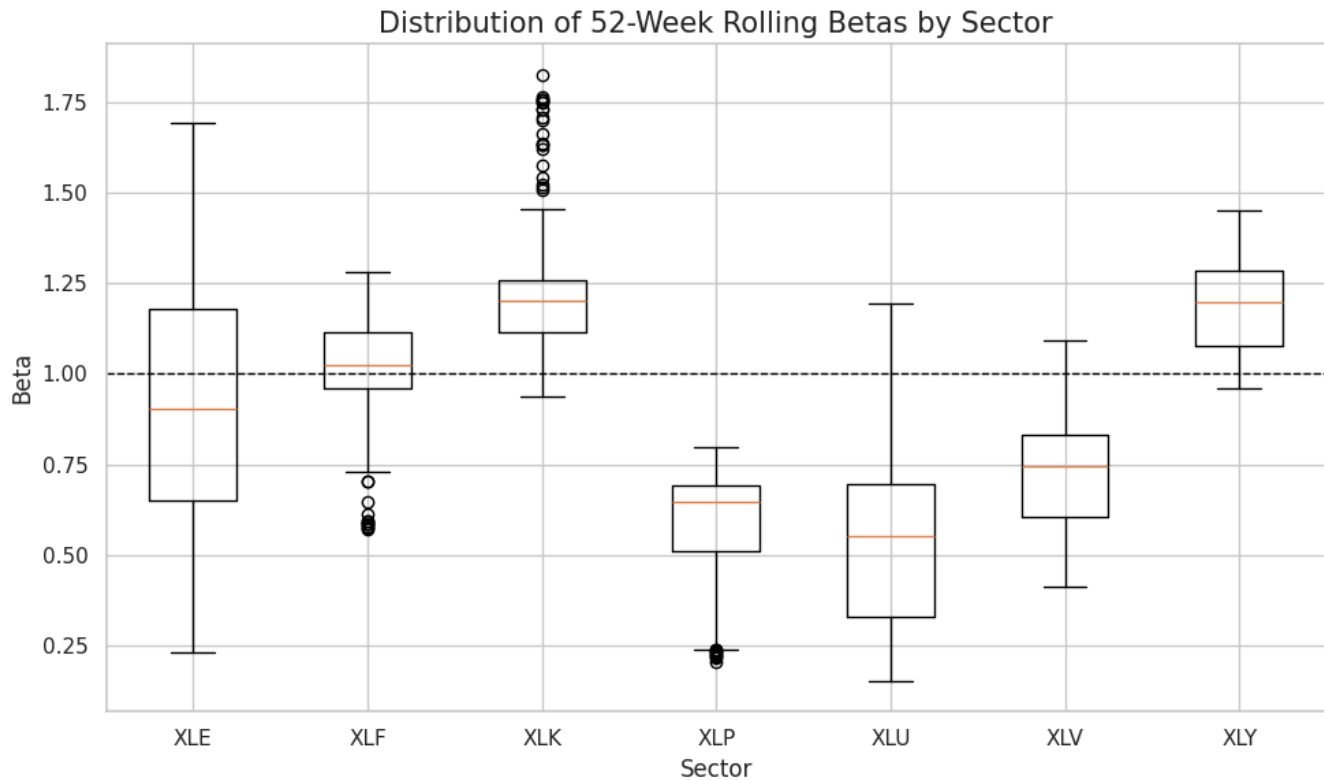


Figure 8. Distribution of 52-week rolling beta by sector.

The boxplots compare the long-run distribution of rolling beta across sectors. XLK and XLY have higher median beta values, while XLP and XLU have lower median beta values. XLK also shows upper-tail beta observations during periods of unusually strong technology-market dependence. In contrast, defensive sectors have tighter beta distributions on average, although they are not completely immune to regime shifts.

The distributional evidence confirms that sector beta differs not only in level but also in stability. A sector can have a low average beta and still experience temporary increases in market sensitivity. For risk management, this distinction is important because investors care about beta behavior during stress periods, not only average behavior across the full sample.

5.3 Average Beta Across Major Regimes

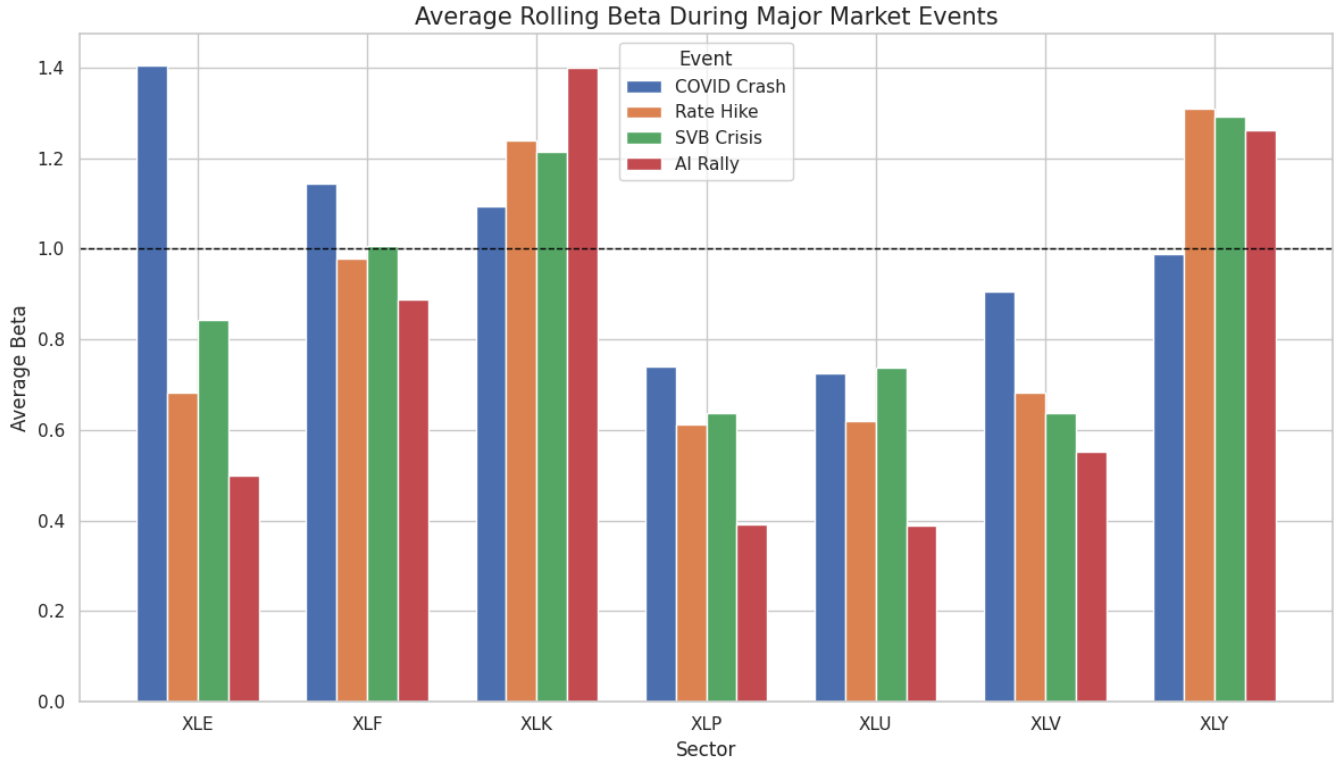


Figure 9. Average rolling beta during major market events.

The event comparison shows that the ranking of sectors changes across regimes. During the COVID crash, energy and financials show elevated market sensitivity, consistent with the broad shock to economic activity and financial conditions. During the Federal Reserve tightening cycle, some traditionally defensive sectors become more sensitive to the market because higher interest rates weaken the appeal of bond-like equity sectors. During the AI-driven rally, technology displays the strongest beta exposure, reflecting the concentration of market performance in large technology stocks.

These findings support the central conclusion of the report: beta should not be viewed as a fixed sector characteristic. A sector can be defensive in one regime and more market sensitive in another. Rolling beta therefore provides more useful information than static labels alone.

6. Predictive Extension: Rolling Beta and Next-Month Volatility

6.1 Motivation and Variable Construction

The rolling beta results show that market exposure varies over time. The next question is whether this variation contains forward-looking information. Specifically, if a sector's beta is high at the end of month t , is its realized volatility higher in month $t + 1$?

The predictor is end-of-month rolling beta, defined as the last available 52-week rolling beta estimate for each sector in each calendar month. The outcome variable is next-month realized volatility, computed as the standard deviation of daily log returns in the following month and scaled by the square root of 21. The predictor and outcome are aligned with a one-month shift to avoid look-ahead bias. The final pooled dataset contains 504 sector-month observations across seven sectors.

$$RV_{i,t+1} = a_i + b_i \text{Beta}_{i,t} + u_{i,t+1}$$

The coefficient b_i measures whether higher current beta is associated with higher next-month volatility. A positive and statistically significant coefficient indicates that rolling beta has predictive content for near-term risk.

6.2 Results

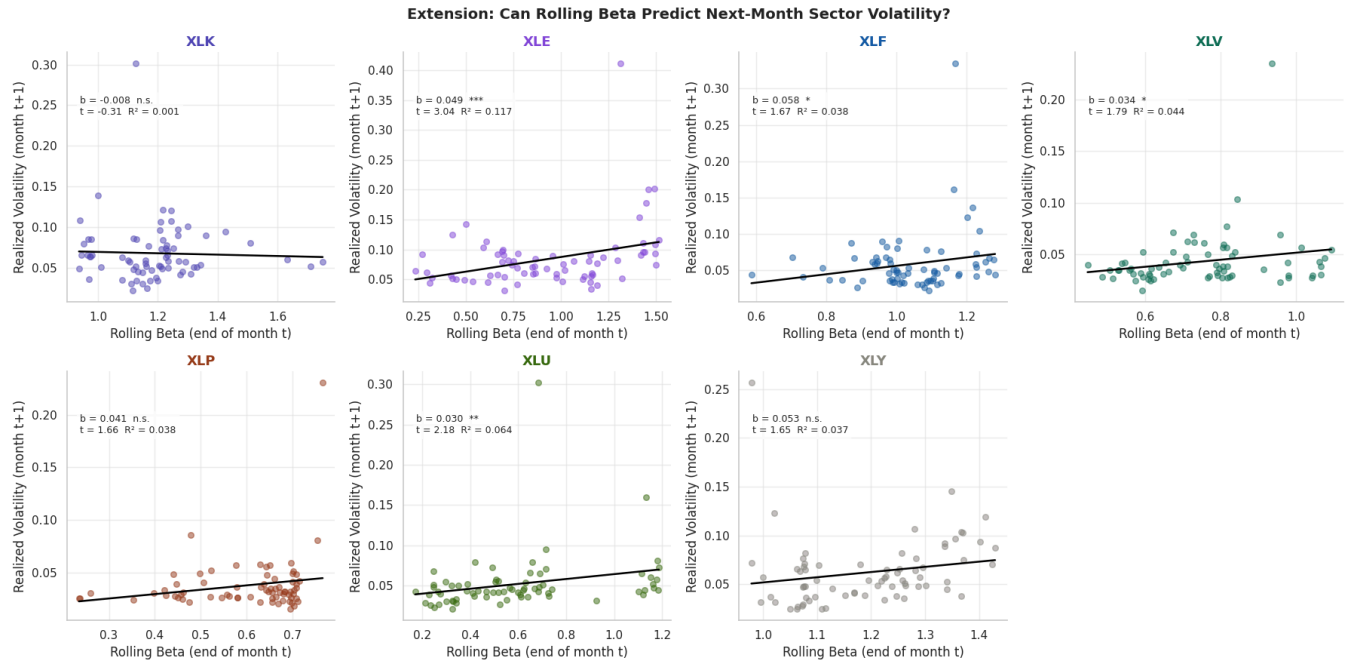


Figure 10. Rolling beta at month t versus realized volatility at month $t + 1$ by sector.

Regression: $\text{realized_vol}_{t+1} = a + b \times \text{beta}_t$ | * $p < 0.10$ ** $p < 0.05$ *** $p < 0.01$

Sector	b coef	t-stat	p-value	R ²	N	Sig
XLK	-0.0084	-0.306	0.7603	0.0013	72	n.s.
XLE	0.049	3.039	0.0033	0.1165	72	***
XLF	0.0577	1.672	0.099	0.0384	72	*
XLV	0.0343	1.793	0.0772	0.0439	72	*
XLP	0.0414	1.656	0.1021	0.0377	72	n.s.
XLU	0.0303	2.179	0.0327	0.0635	72	**
XLY	0.0532	1.652	0.1029	0.0375	72	n.s.
Pooled	0.038	7.202	0.0	0.0936	504	***

Table 4. OLS results for next-month realized volatility regressed on end-of-month rolling beta.

The pooled regression produces a positive and statistically significant coefficient. The estimated slope is 0.038, with a t-statistic of 7.20 and an R-squared of approximately 0.094. Although the R-squared is modest, this is meaningful for a univariate predictive regression using only lagged beta. It suggests that rolling beta captures information about future sector risk beyond a purely descriptive historical summary.

The strength of the relationship varies across sectors. XLE shows the strongest predictive relationship, consistent with energy beta reflecting commodity-related risk shocks. XLU is significant at the 5 percent level, which aligns with the rate-sensitivity of utilities during the tightening cycle. XLF and XLV are marginally significant. XLP and XLY do not reach conventional significance thresholds. XLK is the most important exception: its coefficient is

negative and insignificant. During the AI rally, technology beta rose while realized volatility remained relatively contained, suggesting that higher beta can reflect market leadership rather than higher near-term risk.

6.3 Interpretation and Limitations

The predictive extension should be interpreted as exploratory evidence rather than a complete forecasting model. Rolling beta windows overlap, which may induce serial correlation in the predictor. The pooled regression also combines sectors that may have different volatility dynamics. A more rigorous extension could use Newey-West standard errors, sector fixed effects, or clustered standard errors. Even with this caveat, the results indicate that rolling beta is useful not only for describing historical market exposure but also for assessing near-term risk conditions.

7. Conclusion

This report examines time-varying sector betas for major U.S. sector ETFs from 2018 to 2024. The evidence strongly suggests that sector beta is dynamic rather than fixed. Static CAPM betas provide a useful baseline, but they obscure meaningful shifts in market exposure across regimes.

The analysis produces three main conclusions. First, cyclical sectors such as technology and consumer discretionary generally have higher beta, while consumer staples, health care, and utilities generally have lower beta. Second, defensive behavior is not permanent. During rate-hike and stress regimes, some defensive sectors become more sensitive to market-wide movements. Third, rolling beta has some predictive value for near-term realized volatility in the pooled sample, although the relationship is heterogeneous across sectors and weak for technology during the AI rally.

For portfolio construction and risk management, the implication is clear: sector risk should be evaluated dynamically. Investors should not rely only on full-sample beta estimates or static sector labels. A rolling beta framework provides a more realistic view of how systematic risk moves across sectors during crises, tightening cycles, and market rallies.

Appendix: Reproducibility Notes

The analysis is based on adjusted close prices for sector ETFs and SPY, macroeconomic series from FRED, VIX from Yahoo Finance, manually defined event windows, and rolling CAPM regressions estimated on weekly excess returns. Future revisions should verify that all event-level figures use the same beta source and aggregation method so that tables and charts remain numerically consistent.